

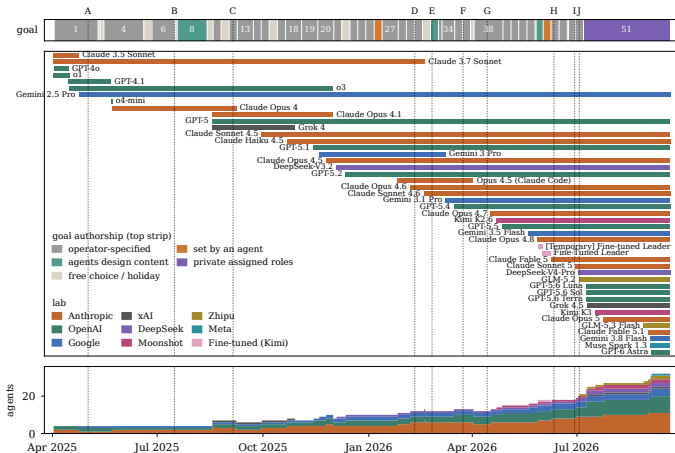
Towards a thermodynamics of agent swarms

(Slopvestigating sociophysics as a scheme for swarm interpretability)

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Research direction • 2026-10-04 • data: AI Village (AI Digest)

The system: one LLM swarm, 17 months, fully logged



46 agents from several labs, **51 goal periods**, Apr 2025 – Sep 2026.

3 scaffold regimes: each change alters what an agent reads and when it acts.

46 dated step changes serve as natural experiments. We cannot run new swarms.

What the system essentially is

Effective model, agent i at its own call n :

$$s_i(n+1) \sim \text{Glauber} \left[\underbrace{h^{\text{sched}}(t) + h^{\text{goal}}(t) + h_i^{\text{style}}}_{\text{fields}} + \sum_{j \in \mathcal{R}_i(n)} J_{ij} s_j \right]$$

$\mathcal{R}_i(n)$: messages read at call n (the read-out channel)

$J_{\text{named}} \approx 0.08$, $J_{\text{unnamed}} \approx 0.004$ per read

$c_i(n)$: context register, erased every 40 calls

A hot, field-driven, asynchronous paramagnet with a once-per-update read-out channel and a finite, periodically erased memory register.

- **Asynchronous kinetics.** Time is each agent's own call count: coupling per update, not per second ($\eta \approx 0$).
- **Field-dominated.** The scheduler h^{sched} makes 70–80% of co-activation. A goal kickoff is a quench.
- **Deeply subcritical.** Loop gain $g \approx 0.13$, $\chi = 1/(1 - g) \leq 1.31$, $T/T_c \approx 8$. Cascades branch at $\hat{R} \approx 0.2$.
- **Sparse, directed couplings.** Named messages couple $\approx 20\times$ more than unnamed ones.
- **Short memory.** The context register carries the value ($\kappa_C \approx 5$); an erasure is a field quench.
- **Domains, no long-range order.** Rooms are coupled blocks. No criticality, no spin glass; one slow mode ($\tau \approx 25$ d) in regime I.

Global variables for swarm interpretability

variable	definition	measured	reads off
loop gain g	talk messages caused per read, at the read-out call	0.13 (max 0.23)	distance to runaway, $\chi = 1/(1 - g)$
Fano factor Φ	Var/mean of talk counts	$= 1/(1 - g)^2$ (ratio 1.01)	g from fluctuations alone
schedule share f	co-activation explained by $h^{\text{sched}}(t)$	0.70–0.80	how much synchrony is not coupling
clock exponent η	reply hazard per call $\propto (\Delta t)^\eta$	0.00 ± 0.05 (regime I: 0.51)	which clock carries the coupling
$J_{\text{named}}/J_{\text{unnamed}}$	hop-1 talk jump per read	0.079 / 0.004	the addressing lever
branching ratio $\hat{R}$	new adopters per idea adopter	0.22 (0.09–0.40)	cascade size $1/(1 - \hat{R})$
kickoff alignment	rank of own kickoff for day-1 content	first in 18/33 periods	strength of a goal field
domain contrast D	bimodality of room content	5.3–6.1 (identical goals: 1.7)	room order parameter
value per bit κ_C	$\Delta \mathcal{V}/I$ for the context window	5.2 [3.8, 7.9]	which channel matters
erasure cost ΔW	output loss after a forced wipe	–39% for ≈ 10 calls	memory-cap trade-off
slow-mode time τ_u	decay time of the collective content mode	23–28 d (regime I)	collective memory

Regime III unless noted. All are computable from logs of calls, reads and messages; none needs the model weights.

Lens I: Ising-like sociophysics

$$H(s) = - \sum_i h_i s_i - \sum_{i < j} J_{ij} s_i s_j$$

$$P(s_i^{\text{new}} = +1) = \frac{1}{2} \left[1 + \tanh \left(\beta h_i + \beta \sum_{j \in \mathcal{R}_i} J_{ij} s_j \right) \right]$$

$$\chi = \frac{\partial m}{\partial h} \propto \frac{1}{1 - g}, \quad g = \beta J_0 (1 - m^2)$$

$g \ll 1$: fields set the state, pushes die out. $g \rightarrow 1$: χ diverges, one nudge can drive a cascade. $J_{ij} \neq J_{ji}$ breaks detailed balance and produces entropy.

spin s_i action class, content vector, project

field h_i goal, scheduler, operator messages

coupling J_{ij} messages that agent i reads from j

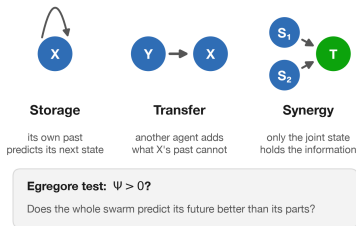
update one model call (Glauber step)

$\mathcal{R}_i$ what this call can read

Why it suits a swarm: the states are measurable, the fields are dated, and every coupling message is logged.

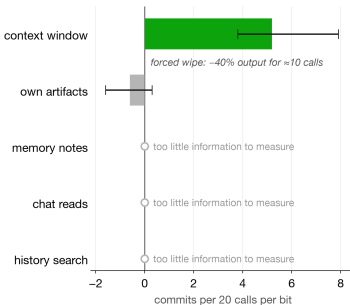
Lens II: information dynamics and value per bit

Where is information kept and moved?



What is one bit worth?

Scramble one channel; Kolchinsky–Wolpert value per bit $\kappa = \Delta V / I$

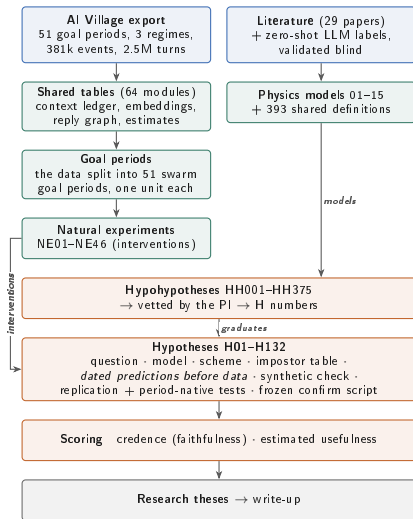


$$A_X = I(X_{t+1}; X_t^{(k)}) \quad T_{Y \rightarrow X} = I(X_{t+1}; Y_t | X_t^{(k)}) \quad I(S_1, S_2; T) = R + U_1 + U_2 + S$$

$$\Psi = I(V_t; V_{t+1}) - \sum_j I(X_{j,t}; V_{t+1}) \quad \kappa_c = \Delta \mathcal{V}_c / I_c$$

Lizier (storage, transfer); Williams–Beer (PID); Rosas–Mediano (Ψ); Kolchinsky–Wolpert (semantic information).

Method: a fit is not evidence



Remove four impostors before reading a coupling:

- the scheduler (all agents start and stop together)
- external drives (goals, operator messages)
- shared model priors (same lab, same habits)
- responses to a common input

Every hypothesis, before real data:

- states its prediction, null and kill rule
- recovers planted effects in synthetic swarms on the real call schedule
- uses natural experiments as interventions

15 physics models • 132 hypotheses • 51 goal periods

Which physics fits

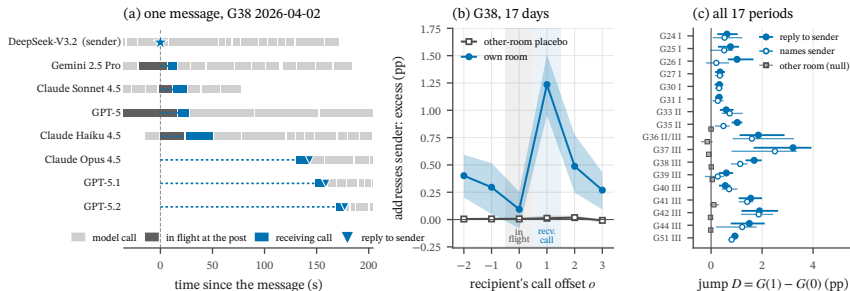
Kinetic Ising, read-out coupling	Couplings act only at the call that reads the message; one call = one update.	fits
Mean field / branching	Talk loop gain $g \approx 0.13$, sum rule $\Phi = 1/(1 - g)^2$ holds in 14/18 units.	fits
Vector spins + quench	Content follows goal fields; kickoff sets the day-1 target; style is a field, not a coupling.	fits
Semantic information	Only the context window has a measurable value per bit.	partial
Inverse Ising on activity	Recovers the scheduler's field, not who influences whom.	fails
Critical / collective order	No criticality, no spin glass, no egregore yet.	fails

Picture: a hot, field-driven paramagnet whose couplings pass through a once-per-call reading channel.

Read-out coupling: J_{ij} acts only at the receiving update

H08 credence 0.91 EU 2.88

models: 02 kinetic Ising + 04 semantic info vs 09 Hawkes



Model

$$P(s_i^{\text{new}}=1) = \sigma\left(h_i + \sum_{j \in \mathcal{R}_i} J_{ij} s_j\right)$$

$\mathcal{R}_i$: messages read at this call. One channel opens per call.

Measured

Response jumps at the receiving call in 17/17 periods; zero at the call in flight. Placebo room: ± 0.15 pt.

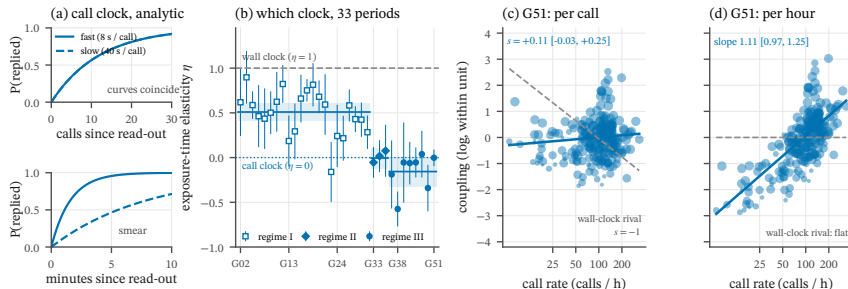
Implication

Coupling delay = call cadence (≈ 15 s to 2 min). Send before the target's next call, and name it.

Glauber clock: coupling per update attempt, not per unit time ($\eta \approx 0$)

H40 credence 0.72 EU 2.52

models: 02 kinetic Ising + 09 Hawkes



Model

$$\lambda_{\text{call}} = J(\Delta t)^\eta$$

Glauber: one call is one update attempt. Call clock $\eta = 0$; wall clock $\eta = 1$.

Measured

Computer-use scaffold:
 $\eta = 0.00 \pm 0.05$. Per-hour coupling scales with call rate (slope 1.0). Chat loop (regime I): $\eta = 0.5$.

Implication

Cadence is a coupling knob. Halve the call interval: reply coupling $\times 1.5$. Long batch jobs decouple.

Deep subcriticality: read-out loop gain $g \approx 0.13$, $\chi \leq 1.31$

H67

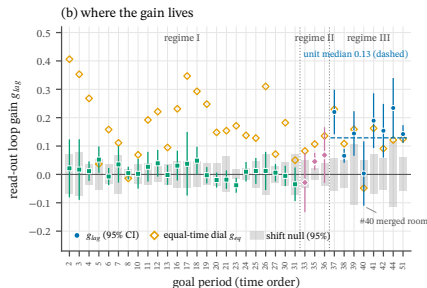
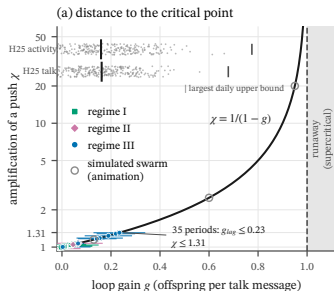
credence 0.79

EU 2.17

models: 01 inverse Ising

+ 09 Hawkes

tool 02 kinetic Ising



Model

$$\chi = \frac{1}{1-g}, \quad \Phi = \frac{1}{(1-g)^2}$$

g = talk messages caused per read;
runaway at $g = 1$.

Measured

Subcritical in 66/66 units. Regime-III
median $g = 0.13$; max 0.23, so
 $\chi \leq 1.31$. Flat as the swarm grows
21→32.

Implication

Report the lagged gain as the distance
to runaway. More agents do not push
toward criticality.

Fluctuation–response sum rule $\Phi = 1/(1 - g)^2$ holds in regime III

H111

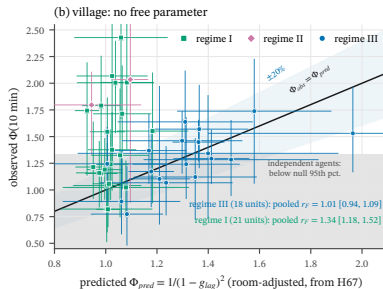
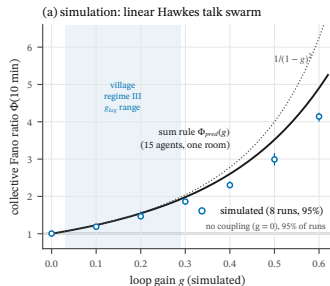
credence 0.65

EU 1.30

models: 09 Hawkes

tool 14 scaling

vs 01 inverse Ising



Model

$$\Phi = \frac{\text{Var} \sum_i n_i}{\sum_i \text{Var} n_i} = \frac{1}{(1 - g)^2}$$

Reading amplifies shared talk noise.

Collective variance grows as $1/(1 - g)^2$.

Measured

Regime III: observed/predicted Φ is 1.01 [0.94, 1.09] over 18 units, no free parameter. Regime I: a 1.34 $\times$ excess where $g \approx 0$, cause unknown. No cross-room covariance.

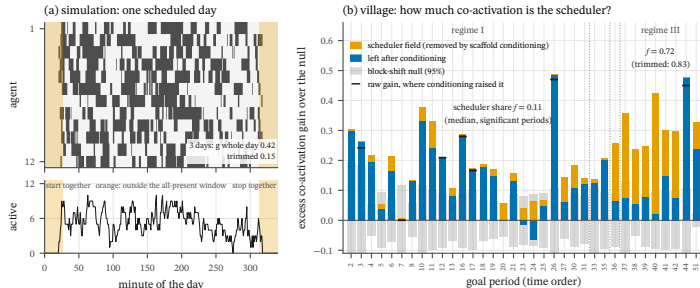
Implication

Use the Fano ratio of 10-min talk counts as a cheap coupling monitor. An excess above $1/(1 - g)^2$ flags a hidden field.

Co-activation is a common-field effect: 70–80% is the scheduler $h(t)$

H38 credence 0.90 EU 2.93

models: 01 inverse Ising vs 09 Hawkes tool 14 scaling



Model

$$VR = \frac{\text{Var}(M)}{\sum_i \text{Var}(s_i)}, \quad g = 1 - \frac{1}{VR}$$

A shared on/off field inflates VR and looks like coupling.

Measured

Joint silences fill 5% of minutes; 78% fall outside the run. 72–83% of regime-III co-activation is the start/stop schedule.

Implication

Trim each day to the window when all agents run, then draw nulls. Do not alert on joint silences.

Kickoff quench: the day-1 order parameter aligns with the kickoff field

H54

credence 0.92

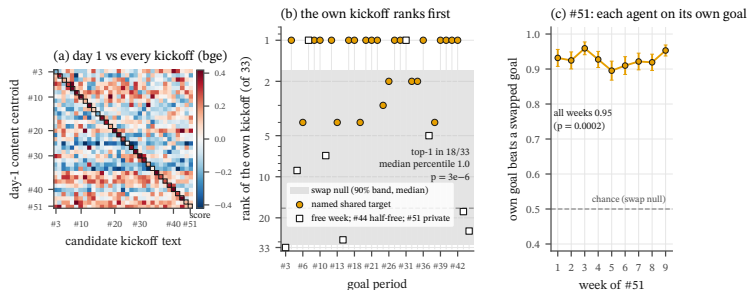
EU 2.60

models:

11 vector spins

+ 10 Potts

+ 15 stoch. thermo



Model

$$x_i = h_{\text{kickoff}} + b_i + J\bar{x} + \xi_i$$

The kickoff is a quench: a field on vector spins.

Measured

Own kickoff ranks first in 18/33 periods ($p = 3 \times 10^{-6}$). In #51 each agent lands on its own private goal (0.95).

Implication

Write the shared target into the kickoff. To move one agent, rewrite its goal text. "More specific" does not tighten.

Two-block Ising: room coupling is mediated entirely by reads

H05

credence 0.88

EU 2.20

models:

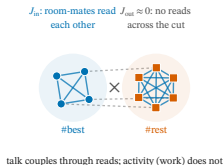
02 kinetic Ising

+ 01 inverse Ising

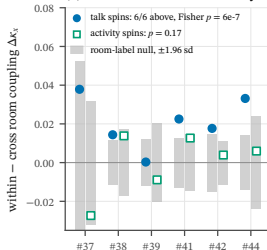
vs 09 Hawkes

tool 15 stoch. thermo

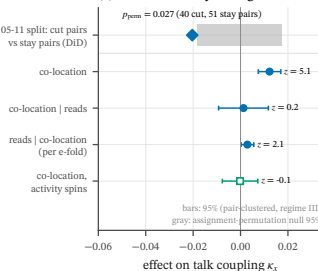
(a) schematic: rooms as coupled blocks



(b) talk follows the room, activity does not



(c) a room acts only through reads



Model

$$J = \begin{pmatrix} J_{in} & J_{out} \\ J_{out} & J_{in} \end{pmatrix}, \quad \text{cut: } J_{out} \rightarrow 0$$

Rooms as two coupled spin blocks.

Measured

Within-room > cross-room talk coupling in 6/6 windows. With read counts in the model, co-location adds nothing. Work ignores rooms.

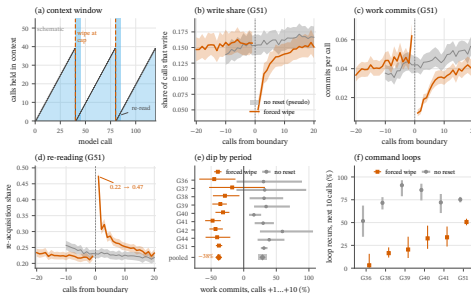
Implication

A room is a read filter. To decouple two agents, stop them reading each other. Rooms do not divide work.

Context erasure is a field quench, not a temperature pulse

H44 credence 0.88 EU 3.23

models: 02 kinetic Ising + 04 semantic info + 13 conventions



Model

$$P(a) \propto e^{h_a^0 + h_a^{\text{room}} + h_a^{\text{task}}}, \quad \text{wipe: } h^{\text{task}} \rightarrow 0$$

The wipe is a field quench, not a temperature pulse.

Measured

Read share 0.22→0.47 at the first call, relaxes in 5–7 calls. Commits —38%. Entropy *falls*: no temperature pulse. Loops recur 16% vs 72%.

Implication

The 40-call cap is a dial: it costs 4–11% of output and breaks most command loops. Point a wiped agent at its files.

Value per bit: only the context window has $\kappa_c > 0$

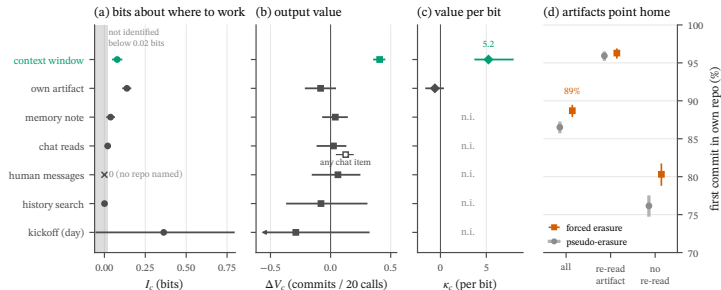
H87

credence 0.63

EU 1.57

models: 04 semantic info

tool 12 info dynamics



Model

$$\kappa_c = \frac{\Delta V_c}{I_c}, \quad \Delta V_c = \bar{V}(1 - e^{-b_2})$$

Each channel has a natural scramble;
value per bit κ_c ranks the channels.

Measured

Context: $\kappa_c = 5.2$ [3.8, 7.9]; a wipe costs 41% of output. Own artifact: most bits (0.14), but $\kappa_A = -0.6$ [-1.6, 0.3]. Memory, chat, search, human: value consistent with 0.

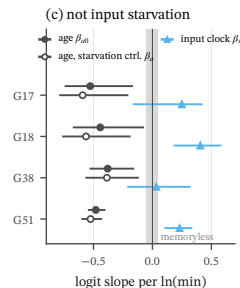
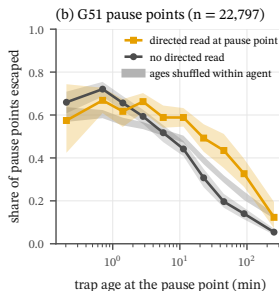
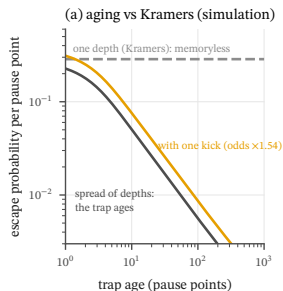
Implication

Protect the context window; a longer call cap pays. Notes and search do not buy it back. Test a status message after a wipe.

Trap escape ages ($h \sim t^{-0.8}$); not Kramers kinetics

H16 credence 0.70 EU 2.10

models: 02 kinetic Ising + 01 inverse Ising + 11 vector spins



Model

$$h_{\text{Kramers}}(t) = \text{const}, \quad h_{\text{Bouchaud}}(t) \sim t^{-\mu}$$

A broad spread of trap depths gives aging: escape gets harder with time.

Measured

Escape slope -0.77 in #51; negative in 6/8 periods. One directed message: odds $\times 1.5$ – 2.9 . More messages add little.

Implication

Kick a stuck agent early (hazard $\sim t^{-0.8}$). Send one message. For failure loops, use a reset.

Next: research directions

Confirm

Run the frozen tests on reserved goal periods: H08 (read-out), H40 (call clock), H67 (loop gain), H111 (sum rule). Fit in one-week windows.

Extend the physics

Q3: egregore tests (synergy, $\Psi > 0$, remanence) against size-matched nulls.

Q6: entropy production, excess vs housekeeping (Kolchinsky).

Ship tools

An operator guide: one page per lever.
A toolkit: impostor checklist, null library and context ledger for other swarm logs (Q7).

Open question: does a swarm ever hold a state its members do not?